A LEAN-VERIFIED SAME-HEIGHT FIBER FRAMEWORK FOR THE RIEMANN HYPOTHESIS

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Abstract. The Riemann Hypothesis asserts that every nontrivial zero of the Riemann zeta function has real part 1/2. This manuscript reports a formal reduction program, implemented and checked in Lean, which reorganizes the problem around the geometry of zero fibers at fixed imaginary height. The central verified equivalence is that RH is equivalent to the assertion that no two critical-strip zeros with the same imaginary part have different real parts. We further formalize energy, cardinality, zero-free-threshold, de la Vallee Poussin 3-4-1, and horizontal critical-point criteria that make this same-height obstruction precise.

The contribution is not a proof of RH. Rather, it is a proof-checked architecture for isolating the exact remaining analytic input: a same-height repulsion or uniqueness principle for zeta zeros. The Lean development contains no active uses of `sorry`, `admit`, or local axioms in the verified surface. We also record a negative result from numerical experimentation: global strict midpoint convexity of $x \mapsto |\zeta(x + i\gamma)|^2$ on the critical strip is too strong and is contradicted already at small height. The framework therefore narrows the next viable target to local horizontal curve/argument-principle constraints rather than global convexity.

1. Introduction

Riemann's 1859 memoir connected the distribution of primes with the zeros of the analytically continued zeta function and formulated what is now called the Riemann Hypothesis [Rie59]. In modern notation, the nontrivial zeros are those in the critical strip

$$\mathcal{S} = \{s \in \mathbb{C} : 0 < \mathrm{Re}(s) < 1\},$$

and RH asserts that every such zero satisfies $\mathrm{Re}(s) = 1/2$. The problem has been attacked through zero-free regions, explicit formulae, positivity criteria, spectral interpretations, random matrix statistics, and rigorous computation; see, for example, Titchmarsh and Heath-Brown [THB86], Edwards [Edw74], Iwaniec and Kowalski [IK04], the DLMF survey [NIS26], and recent formalization work in Lean [LS25].

This paper describes a complementary proof-search framework. Instead of trying to estimate the entire critical strip directly, we organize possible

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counterexamples by their imaginary height. For each $\gamma \in \mathbb{R}$, let

$$\mathcal{F}_\gamma = \{s \in \mathbb{C} : \zeta(s) = 0,\ 0 < \mathrm{Re}(s) < 1,\ \mathrm{Im}(s) = \gamma\}.$$

The key observation is elementary once the usual zeta symmetries are available: an off-line zero forces a second zero at the same height, reflected across the critical line. Consequently, RH is equivalent to real-coordinate uniqueness in every fixed-height zero fiber. The value of this reformulation is not that it makes RH easier for free. The value is that it makes the missing analytic ingredient exact: rule out same-height collisions.

Contributions. The verified development makes the following contributions.

(i) It gives Lean-checked equivalences between RH and same-height zero uniqueness, no-collision formulations, and fiberwise real-coordinate uniqueness.
(ii) It defines a finite fiber energy and verifies that vanishing of this energy at every height is equivalent to RH.
(iii) It proves that a cardinality bound $|\mathcal{F}_\gamma| \le 1$ for every γ is sufficient for RH, using formalized conjugate symmetry and centroid identities.
(iv) It records a zero-free-threshold equivalence: RH is equivalent to zero-freeness to the right of every vertical line $\mathrm{Re}(s) = b$ with $b > 1/2$.
(v) It formalizes a de la Vallee Poussin 3-4-1 Euler-product bridge in the half-plane $\sigma > 1$, including the lower bound

$$|\zeta(\sigma)^3\zeta(\sigma + it)^4\zeta(\sigma + 2it)| \ge 1.$$

(vi) It tests and rejects a natural global convexity route, then replaces it by sharper conditional horizontal critical-point criteria.

All statements above refer to the currently checked Lean files in the `Reinmann/` directory. The project intentionally distinguishes verified Lean modules from research notes and numerical experiments.

Status of the result. This manuscript should not be read as a claimed solution of RH. The remaining gap is the following analytic principle.

> *Same-height repulsion problem.* Prove that two distinct zeros of $\zeta(s)$ in the open critical strip cannot have the same imaginary part.

The formal development proves that this principle is exactly strong enough to settle RH. It does not prove the principle.

2. Literature review and novelty

2.1. **Classical zero-free methods.** The classical proof of the prime number theorem by Hadamard and de la Vallee Poussin established, among other things, nonvanishing on the line $\mathrm{Re}(s) = 1$. The DLMF records this boundary zero-free result and the standard symmetries of the critical strip [NIS26]. This line of work developed into explicit zero-free regions and bounds for

$\zeta(s)$, as systematized in Titchmarsh–Heath-Brown [THB86] and later analytic number theory texts [IK04].

Our work uses the classical boundary line as a formal component, but it does not try to improve explicit zero-free regions. The zero-free-threshold theorem in Section 5 instead records a logical equivalence: if all vertical strips $\mathrm{Re}(s) > b$, $b > 1/2$, are zero-free inside the critical strip, then RH follows, and conversely RH gives these zero-free statements.

2.2. **Equivalent positivity criteria.** Several well-known criteria re-express RH as a positivity assertion. Weil's explicit formula gives positivity criteria for test functions [Wei52]. Li's criterion states that RH is equivalent to nonnegativity of a sequence of coefficients derived from the xi function [Li97], and Bombieri–Lagarias generalized this viewpoint [BL99]. These criteria are close in spirit to the present work because they replace the location statement by another global assertion whose failure detects an off-line zero.

The difference is the granularity of the obstruction. Li and Weil criteria aggregate over all zeros. The present energy criterion is fiberwise: each imaginary height γ receives an energy measuring the squared distance of zeros in $\mathcal{F}_\gamma$ from the critical line. RH is equivalent to the vanishing of this entire family of local energies.

2.3. **Spectral and physical approaches.** The Hilbert–Polya idea asks for a self-adjoint operator whose spectrum encodes the imaginary parts of zeta zeros. Berry and Keating related the smooth zero counting law to quantizations of $H = xp$ [BK99]; Connes interpreted critical zeros as an absorption spectrum in a noncommutative geometric trace formula [Con99]. These programs are deep and structural, but a complete RH proof would still need a mechanism forcing critical-line location.

The same-height framework is compatible with spectral intuition but does not postulate an operator. It isolates a simpler spectral-looking obstruction: degeneracy of the imaginary height map on the set of strip zeros. A Hilbert–Polya proof would imply such nondegeneracy after identifying the correct self-adjoint model, but the present framework leaves the analytic source of the repulsion open.

2.4. **Random matrix statistics and simplicity.** Montgomery's pair-correlation program and Odlyzko's computations connect local zero statistics with eigenvalues of random Hermitian matrices [Mon73, Odl87]. Recent work has continued to sharpen what can be proved about pair correlation and simple zeros; for instance, Baluyot–Goldston–Suriajaya–Turnage-Butterbaugh prove an unconditional Montgomery-type theorem with a conditional application to the proportion of simple zeros in a thin box near the critical line [BGSTB24].

These results concern vertical spacings and multiplicity. Same-height uniqueness is a different obstruction. A multiple zero on the critical line is not automatically a same-height collision with different real parts, and a

same-height off-line pair could consist of two simple zeros. Thus the present formulation is not subsumed by the usual simplicity conjecture, although both are zero-repulsion statements.

2.5. **Rigorous computation and formal proof.** Rigorous numerical methods have verified RH up to finite heights; Platt isolated nontrivial zeros up to height 30,610,046,000 with rigorous error bounds [Pla17]. Formal proof assistants address a different problem: they remove ambiguity from symbolic arguments. The Lean mathematical library provides the foundation used here [The20], and recent work formalizes zeta and Dirichlet L-functions, including a formal statement of RH [LS25].

The novelty of this paper lies at the intersection of these directions: proof-checked reductions, expressed in Lean, that turn RH into a precise same-height uniqueness target. The contribution is not a new numerical verification range and not a new zero-free region; it is a verified logical spine for future analytic work.

3. Formal setup

We use the Riemann zeta function $\zeta(s)$, meromorphically continued to $\mathbb{C}$, and let RH denote the standard statement that all nontrivial zeros lie on the line $\mathrm{Re}(s) = 1/2$. The Lean development imports mathlib's `RiemannHypothesis : Prop`; local modules then prove reductions to this formal target.

Definition 3.1 (Critical-strip fiber)**.** *For $\gamma \in \mathbb{R}$, define*

$$\mathcal{F}_\gamma = \{s \in \mathbb{C} : \zeta(s) = 0,\ 0 < \mathrm{Re}(s) < 1,\ \mathrm{Im}(s) = \gamma\}.$$

Definition 3.2 (Fiber uniqueness)**.** *The zero-fiber uniqueness property is*

$$\forall \gamma \in \mathbb{R}, \quad \forall s, t \in \mathcal{F}_\gamma, \quad \mathrm{Re}(s) = \mathrm{Re}(t).$$

Equivalently, no two critical-strip zeros of ζ have the same imaginary part and different real parts.

Definition 3.3 (Fiber energy)**.** *For a finite fiber, define*

$$E(\gamma) = \sum_{s \in \mathcal{F}_\gamma} (\mathrm{Re}(s) - 1/2)^2.$$

The Lean implementation uses a finite set representation supplied by the local zero-finiteness module.

Remark 3.4. *The finiteness of each fiber is not an RH assumption. It follows from isolation of zeros of a nonzero analytic function in the relevant bounded horizontal slice; in the repository this is encapsulated by the module `ZeroFiberFiniteness.lean`.*

4. Verified main reductions

4.1. **Same-height uniqueness.**

Theorem 4.1 (Fiber uniqueness criterion)**.** *RH is equivalent to zero-fiber real-coordinate uniqueness at every imaginary height.*

Lean status. This is formalized by

`riemannHypothesis_iff_forall_zeroFiberRealUnique_unconditional`

in `RiemannHypothesisReduction.lean`. Earlier versions treated strip conjugation as an obligation; the current development closes this bridge using the conjugation-symmetry module. □

Remark 4.2. *The theorem explains why the same-height target is not a weakening of RH. It is an equivalent form. Any proposed proof of same-height uniqueness therefore needs analytic strength comparable to a proof of RH.*

4.2. **Energy criterion.**

Theorem 4.3 (Fiber-energy criterion)**.** *RH is equivalent to*

$$\forall \gamma \in \mathbb{R}, \quad E(\gamma) = 0.$$

Lean status. This is formalized as

`riemannHypothesis_iff_forall_fiberEnergy_zero`

in `SpectralEnergyCriterion.lean`. The proof uses nonnegativity of each summand and the fact that the squared real displacement vanishes exactly on the critical line. □

Corollary 4.4 (Positive energy detects RH failure)**.** *RH fails if and only if some height has positive fiber energy.*

Lean status. This is formalized as

`not_riemannHypothesis_iff_exists_fiberEnergy_pos`

in `FiberEnergyGap.lean`. □

4.3. **Centroid and cardinality.** Conjugate/functional-equation symmetries force off-line zeros to occur in mirror pairs across $\mathrm{Re}(s) = 1/2$. This gives a useful centroid identity.

Theorem 4.5 (Fiber centroid)**.** *If $\mathcal{F}_\gamma$ is nonempty, then the average of the real parts of its zeros is* $1/2$.

Lean status. The finite-sum identity is formalized by

`fiber_centroid_eq_half`

in `FiberCentroid.lean`. □

Corollary 4.6 (Cardinality sufficient condition)**.** *If $|\mathcal{F}_\gamma| \leq 1$ for every γ, then RH holds.*

Lean status. The proof is formalized by

`riemannHypothesis_of_all_fiber_card_le_one`

in `FiberCentroid.lean`. A one-point fiber has centroid $1/2$, so its only point lies on the critical line; empty fibers contribute nothing. □

5. Zero-free and de la Vallee Poussin bridges

5.1. **Vertical threshold formulation.**

Definition 5.1. *For $b \in \mathbb{R}$, let* ZeroFreeRightOf(b) *denote the assertion that no critical-strip zero satisfies* $\mathrm{Re}(s) > b$.

Theorem 5.2 (Threshold criterion)**.** *RH is equivalent to*

$$\forall b > 1/2, \quad \text{ZeroFreeRightOf}(b).$$

Lean status. This is formalized by

`riemannHypothesis_iff_zeroFreeRightOf_all_gt_half`

in `ZeroFreeEngine.lean`. □

5.2. **The 3-4-1 product.**

The de la Vallee Poussin boundary-line proof uses the elementary trigonometric positivity

$$3 + 4\cos\theta + \cos(2\theta) = 2(1 + \cos\theta)^2 \geq 0.$$

In the half-plane $\sigma > 1$, this positivity yields an Euler-product lower bound.

Theorem 5.3 (3-4-1 zeta product)**.** *For $\sigma > 1$ and $t \in \mathbb{R}$,*

$$\left|\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it)\right| \geq 1.$$

Lean status. The trigonometric component is formalized in `ZeroFreeEngine.lean`. The Euler-product bridge and the zeta-specialized product bound are formalized by

`zeta_341_modulus`

in `Zeta341GlobalBridge.lean`. The conditional limit contradiction against a boundary zero is recorded in `VdpConditional.lean`. □

Remark 5.4. *This component is a verified bridge, not a new proof of the classical boundary zero-free theorem. Mathlib already supplies the boundary nonvanishing fact needed by the active Lean development.*

6. Horizontal repulsion experiments

The same-height formulation suggests a horizontal repulsion principle: for fixed γ, the function $x \mapsto \zeta(x + i\gamma)$ should be unable to hit zero twice in $0 < x < 1$. We formalized two classes of sufficient conditions.

6.1. **A failed convexity target.** A natural first target was strict midpoint convexity of

$$x \mapsto |\zeta(x + i\gamma)|^2$$

on the critical strip. Lean verifies that this global property would imply RH:

`riemannHypothesis_of_horizontalStrictMidpointConvex`.

However, numerical testing falsifies the proposed global convexity. With $\gamma = 0.5$, $x = 0.66$, $y = 0.98$, and $m = 0.82$, high-precision evaluation gives

$$|\zeta(m + i\gamma)|^2 \approx 3.0092902087845057592,$$

whereas

$$\frac{|\zeta(x + i\gamma)|^2 + |\zeta(y + i\gamma)|^2}{2} \approx 2.9997768238141877056.$$

Thus the midpoint value is larger than the endpoint average. The convexity route is therefore too strong.

6.2. **Horizontal critical-point criteria.** The failed convexity experiment motivates a local alternative. If a real-valued differentiable function vanishes at two points, Rolle's theorem forces a critical point between them. For the complex-valued horizontal slice

$$f_\gamma(x) = \zeta(x + i\gamma),$$

one should apply this idea componentwise to $\operatorname{Re} f_\gamma$ and $\operatorname{Im} f_\gamma$. The Lean development therefore formalizes a conditional criterion: if two distinct same-height zeros forced a forbidden pair of componentwise critical points between them, then every fiber has cardinality at most one, and RH follows.

Theorem 6.1 (Componentwise horizontal critical criterion)**.** *Assume the componentwise Rolle principle for horizontal zeta slices and assume that no two distinct zeros in one fiber admit the resulting forbidden componentwise critical pair. Then RH holds.*

Lean status. The theorem is formalized by

`riemannHypothesis_of_componentwiseCriticalExclusion`

in `HorizontalCritical.lean`. □

Remark 6.2. *The conditional critical-exclusion criterion does not establish its own hypothesis. Componentwise Rolle points need not coincide, and a same-height pair does not by itself force a zero of the complex derivative on the horizontal segment. No checked theorem in this paper supplies the missing zeta-specific exclusion estimate.*

7. Status of the proposed phase hypothesis

Write $\theta'(\sigma) = \mathrm{Im}(\zeta'/\zeta)(\sigma + i\gamma)$ where the denominator is nonzero. The proposed hypothesis (H_θ) asks for an appropriate nonvanishing or sign bound on these horizontal segments. It remains an analytic research target. A proof that this bound implies componentwise critical exclusion would itself require a valid zeta-specific argument; no such bridge is established here. We do not claim that Lean has proved an equivalence between this phase hypothesis and RH.

An earlier draft asserted that two zeros in a thin rectangle force exactly one derivative zero there and hence a horizontal phase crossing. This is not a general argument-principle consequence. For example,

$$f(z) = e^{iz}z(z-1), \qquad \mathrm{Im}\,\frac{f'(x)}{f(x)} = 1 \quad (0 < x < 1),$$

has two zeros at the endpoints and no phase-derivative zero between them. The assertion $N(f') = N(f) - 1$ on an arbitrary thin rectangle also needs additional hypotheses. This example does not settle a zeta-specific question; it identifies a missing implication in the proposed argument.

Zero-product formulas must specify their convergence, constants and archimedean terms. Finite phase samples or finite zero sums give no uniform nonvanishing bound without a controlled remainder. GUE or pair-correlation statistics do not imply a pointwise prohibition on a cancellation at every height. The earlier numerical constants are therefore not used as rigorous analytic bounds, and the preceding formal reductions remain conditional at precisely their named exclusion hypotheses.

8. Unproved operator and flow proposals

The files `SpectralPhaseBounds.lean.draft`, `TopologicalFlow.lean.draft` and `DynamicalRepulsion.lean.draft` record proposals outside the active Lean proof surface. They do not prove the phase hypothesis or RH.

For the first-order expression $-i\,d/d\sigma + \theta'(\sigma)$, self-adjointness requires a specified operator domain and a proved boundary condition analysis. Conjugation symmetry alone does not supply it, and a zero of a real potential does not force nonreal spectrum. Likewise, a winding count for a function is not a saddle-point theorem for its derivative. The proposed heat-flow argument needs a correctly normalized evolution, control of the infinite interaction sum and collisions, and a justified limiting statement. None of these obligations is discharged by the displayed heuristics in earlier drafts.

The defensible outputs are finite calibration and formalization of stated reductions. The universal Toeplitz ladder, all-window Weil positivity and Green-form paired budget retain RH-level obligations. In particular, `XiGreenModel.lean` contains `green_paired_budget_iff_riemannHypothesis`; the Toeplitz equivalence in `RHReductionCapstone.lean` explicitly takes classical bridge hypotheses. These distinctions must be retained when comparing tracks.

Neither finite experiments nor a new presentation of an RH-level hypothesis constitutes a route to RH by itself.

9. Verification methodology

The Lean repository uses a strict verification harness, `safe-verify.sh`. Before invoking Lean, the harness scans active Lean files for forbidden proof shortcuts including `sorry`, `admit`, `axiom`, `constant`, `opaque`, and `unsafe`. It then runs the Lean toolchain on the active files. Research notes and draft files are kept outside the active proof surface.

This verification method does not itself prove mathematical novelty. It does, however, give a precise audit boundary: the named theorems in the active Lean surface typecheck without local proof placeholders. The analytic gap is not hidden in Lean; it is explicitly named as a conditional hypothesis when it has not been proved.

10. Discussion

The same-height framework contributes a structural diagnostic for RH proof search. It converts an off-line-zero problem into a degeneracy problem for the height projection

$$s \mapsto \mathrm{Im}(s)$$

restricted to critical-strip zeros. This view aligns with spectral intuition: if zeta zeros were realized as a genuinely one-dimensional self-adjoint spectrum, then same-height collisions off the critical line would be geometrically unnatural. But the framework does not assume such a spectral model.

The most important limitation is also the main research target. No currently formalized theorem proves same-height repulsion for zeta zeros. Statistical spacing laws, finite computation, and known proportions of simple zeros provide evidence for zero repulsion, but they do not supply the deterministic fiberwise uniqueness statement needed here. The failed convexity experiment is therefore useful: it rules out an attractive but false sufficient condition and pushes the program toward local horizontal argument-principle methods.

11. Conclusion

The paper presents a Lean-verified reduction framework for RH centered on same-height zero fibers. The central equivalence says that RH is exactly the absence of same-height critical-strip zero collisions with different real parts. Additional verified criteria express the same obstruction through fiber energy, fiber cardinality, vertical zero-free thresholds, and conditional horizontal critical-point exclusions. The development also formalizes a 3-4-1 de la Vallee Poussin product bridge in the half-plane $\sigma > 1$.

The next mathematical step is sharply defined: prove a local same-height repulsion principle for $\zeta(x + i\gamma)$. A plausible direction is a horizontal

argument-principle or componentwise Rolle theorem strengthened by zeta-specific phase information. Until that principle is proved, the work is a verified reduction and proof-search architecture, not a proof of RH.

Acknowledgements

The Lean development depends on mathlib and on its formalization of the Riemann zeta function and the Riemann Hypothesis statement.

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Appendix A. Lean theorem map

Mathematical statement	Lean theorem/module
RH iff same-height uniqueness	`riemannHypothesis_iff_forall_zeroFiberRealUnique_unconditional`, `RiemannHypothesisReduction.lean`
RH iff no same-height collision	`riemannHypothesis_iff_noSameImaginaryPartCollision_unconditional`, `RiemannHypothesisReduction.lean`
RH iff all fiber energies vanish	`riemannHypothesis_iff_forall_fiberEnergy_zero`, `SpectralEnergyCriterion.lean`
Positive fiber energy detects RH failure	`not_riemannHypothesis_iff_exists_fiberEnergy_pos`, `FiberEnergyGap.lean`
Fiber centroid equals 1/2	`fiber_centroid_eq_half`, `FiberCentroid.lean`
Fiber cardinality ≤ 1 implies RH	`riemannHypothesis_of_all_fiber_card_le_one`, `FiberCentroid.lean`
Threshold zero-free equivalence	`riemannHypothesis_iff_zeroFreeRightOf_all_gt_half`, `ZeroFreeEngine.lean`
3-4-1 zeta product lower bound	`zeta_341_modulus`, `Zeta341GlobalBridge.lean`
Convexity sufficient condition	`riemannHypothesis_of_horizontalStrictMidpointConvex`, `HorizontalRepulsion.lean`
Componentwise critical exclusion sufficient condition	`riemannHypothesis_of_componentwiseCriticalExclusion`, `HorizontalCritical.lean`
Unproved spectral proposal (inactive draft)	`SpectralPhaseBounds.lean.draft`
Unproved topological proposal (inactive draft)	`TopologicalFlow.lean.draft`
Unproved dynamical proposal (inactive draft)	`DynamicalRepulsion.lean.draft`

Appendix B. Submission note

For peer review, the accurate submission category is a formalization or proof-assistant-assisted reduction paper, not a claimed proof of the Riemann Hypothesis. A suitable arXiv classification would be `math.NT` with cross-listing to `cs.LO` if the Lean development is included as a reproducible artifact.

Independent researcher